

AWS Generative AI Services

A Concise Overview for Builders & Decision Makers

■ 1. Amazon Bedrock

What It Is

- Fully managed service to build and scale generative AI applications.
- Provides access to high-performing foundation models (FMs) from leading AI companies via a single API.

Key Features

- Model choices: Amazon Titan, Anthropic Claude, AI21 Labs, Cohere, Meta Llama, Stability AI, Mistral.
- Serverless — no infrastructure to manage.
- Fine-tuning & customisation of FMs with your own data.
- Built-in security, privacy, and compliance (data never leaves your AWS account).
- Native integration with AWS services: S3, Lambda, CloudWatch, IAM, etc.

Common Use Cases

- Chatbots and virtual assistants.
- Text summarisation, classification, and generation.
- Code generation and explanation.
- Image generation (Stability AI / Titan Image models).

■ 2. Amazon Q Developer

What It Is

- AI-powered developer assistant built specifically for software development tasks.
- Deeply integrated with AWS — understands your AWS environment, docs, and best practices.

Key Features

- Inline code suggestions in VS Code, JetBrains, and other IDEs.
- Explains, refactors, debugs, and documents code on demand.
- Security vulnerability scanning with auto-remediation suggestions.
- AWS Console chat — ask natural-language questions about your AWS resources.
- Upgrade assistant: automates Java version migrations and framework upgrades.

Common Use Cases

- Accelerating development with AI pair-programming.
- Automating repetitive coding tasks and boilerplate.
- Learning AWS services through contextual Q&A.
- Identifying and fixing security issues early in the SDLC.

■ 3. RAG — Retrieval-Augmented Generation

What It Is

- A technique that enhances LLM responses by retrieving relevant, up-to-date information from an external knowledge base before generating an answer.
- Reduces hallucinations and keeps answers grounded in your own data.

How It Works (3-step flow)

1. Retrieve — A query is embedded and matched against a vector store to fetch relevant document chunks.
2. Augment — Retrieved context is injected into the prompt alongside the user's question.
3. Generate — The LLM produces a response grounded in the retrieved context.

AWS Implementation — Knowledge Bases for Bedrock

- Managed RAG pipeline: ingest documents from S3, auto-chunk, embed, and store in a vector DB.
- Supported vector stores: OpenSearch Serverless, Pinecone, Aurora PostgreSQL (pgvector), Redis Enterprise.
- Automatic sync keeps the knowledge base current as source data changes.

Common Use Cases

- Enterprise Q&A bots over internal documentation.
- Customer support with product/policy knowledge.
- Research assistants grounded in proprietary datasets.

■ 4. AI Agents in Amazon Bedrock

What It Is

- Fully managed service to create autonomous agents that can plan, reason, and execute multi-step tasks.
- Agents decide which actions to take and in what order — without hardcoded workflows.

Key Components

- Foundation Model — the reasoning engine (choose from Bedrock's model catalogue).
- Action Groups — define what the agent CAN do (API calls, Lambda functions).
- Knowledge Bases — optional RAG integration for domain-specific knowledge.
- Orchestration — the agent automatically manages the chain of thought and tool calls.

Agent Collaboration

- Multi-agent systems: a supervisor agent delegates sub-tasks to specialised agents.
- Inline agents: create dynamic, session-scoped agents programmatically at runtime.

Common Use Cases

- Automated IT operations (create tickets, query systems, trigger remediation).
- Agentic customer service (look up orders, process returns, update records).
- Data analysis pipelines driven by natural language.

■ 5. Amazon SageMaker

What It Is

- AWS's comprehensive, end-to-end ML platform for building, training, and deploying machine learning models at scale.
- Supports classical ML, deep learning, and large language model (LLM) workflows.

Core Modules

- SageMaker Studio — unified IDE for the entire ML lifecycle.
- SageMaker JumpStart — 1-click deployment of pre-built FMs and solution templates.
- SageMaker Training — distributed training on managed GPU/CPU clusters.
- SageMaker Pipelines — CI/CD for ML workflows (MLOps).
- SageMaker Inference — real-time, batch, and serverless endpoints.
- SageMaker Clarify — bias detection and model explainability.

Generative AI Focus

- Fine-tune open-source LLMs (Llama, Mistral, Falcon) on custom data.
- Deploy and host your own models alongside Bedrock-managed FMs.
- SageMaker HyperPod: purpose-built distributed training clusters for large models.

Common Use Cases

- Custom model training and fine-tuning at scale.
- MLOps pipelines from experimentation to production.
- Hosting private or open-source LLMs with full control.

Quick Comparison

Service	Category	Best For	Skill Level
Amazon Bedrock	Managed AI Platform	FM access & customisation	Low – High
Amazon Q Developer	AI Dev Assistant	Code generation & AWS help	Low
RAG (Bedrock KB)	AI Pattern / Technique	Grounded Q&A; over your data	Medium
AI Agents (Bedrock)	Agentic AI	Autonomous multi-step tasks	Medium – High
Amazon SageMaker	Full ML Platform	Custom model training & MLOps	Medium – High